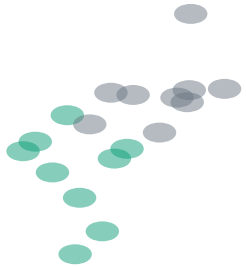
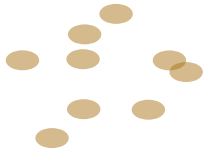
HEPTOKENS · PARTICLE-LIBRARY TOKENIZER · REAL CMS DATA

The Particle Library



A physics-defined token for AI-ready HEP datasets — five campaigns, one criterion.





THE PROBLEM

Ship tokens, not particle-flow candidates

CMS and ATLAS reconstruct particles that outside groups can't access — yet those groups want to train foundation models on the events. Publish a per-constituent token as the shareable artifact and the consumer never needs the reconstruction.

The token is the release interface — versioned by dimension, preprocessing hash, and checkpoint.

THE IDEA

Define the metric in physics, not augmentations

> Same PID, same charge, similar pT ($\Delta pT \leq 10\%$) and same detector region ($|\Delta\eta|, |\Delta\varphi| \leq 0.1$) → positives. Everything else is pushed away.

Explicit, auditable, versionable — and computable on any experiment's particles, so cross-detector alignment can fall out of mixed batches.

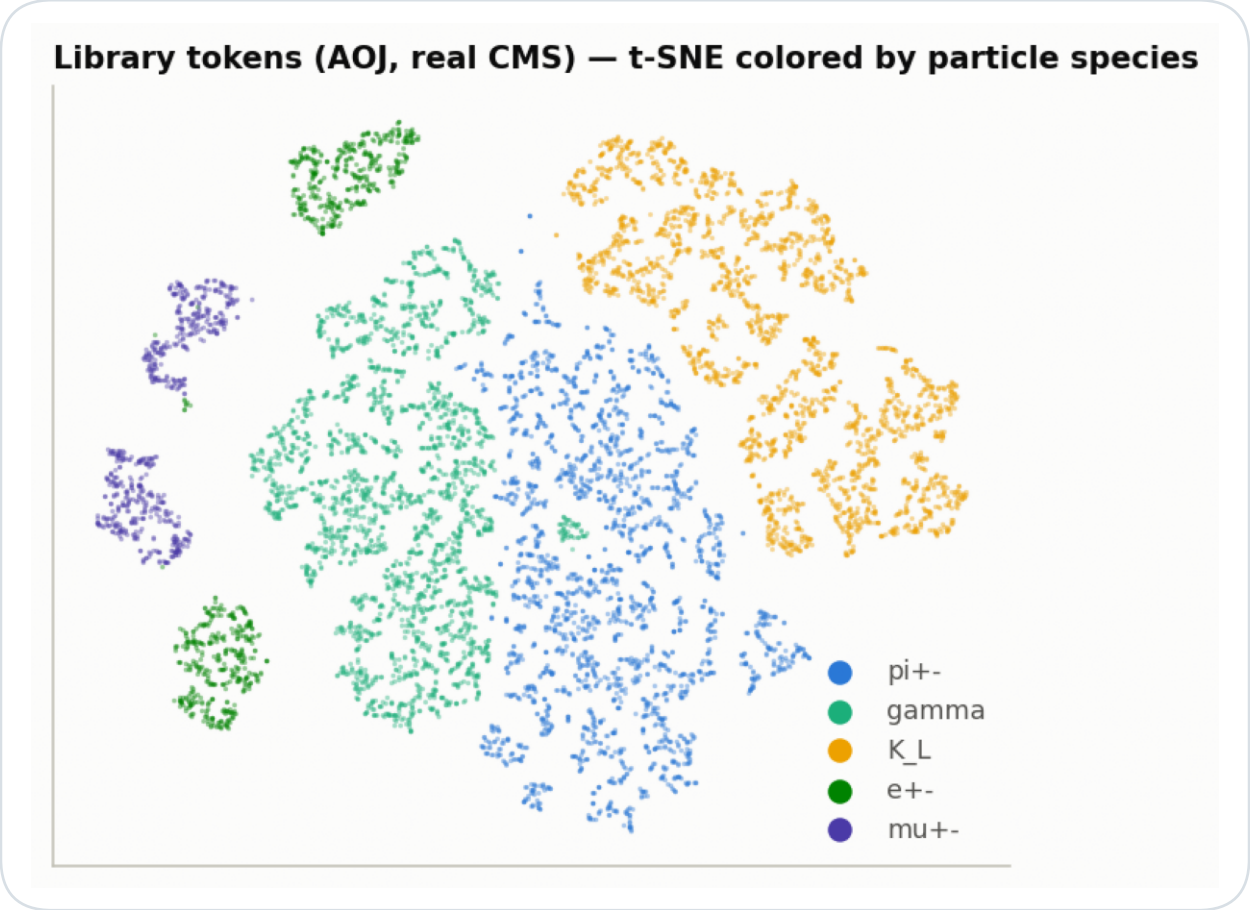
CAMPAIGN 1 · ACHIEVED

The criterion is achieved on real CMS data

0.992

criterion recall@8 — of an anchor's available same-PID/charge/pT/region partners, the fraction ranked in its top-8. A 20× reduction in retrieval misses vs the untrained hand-crafted null.

CAMPAIGN 1 · SPECIES MAP



t-SNE of library tokens, real CMS. π, γ, K_L, e, μ separate cleanly with kinematic substructure inside each island — the species structure is metric, not linear.

CAMPAIGN 1 · THE PIVOTAL CALL

Context-free was decisive

context transformer 0.052

→ **MLP 0.9992**

A jet-context encoder makes the same particle embed differently in different events — which destroys the library property that cross-experiment alignment needs. The token must be a function of one particle.

CAMPAIGN 2 · LIBRARY VS VQ-VAE

Complementary, not competing

PHYSICS LIBRARY

owns the criterion

- recall@8 0.9991
- ~80× fewer retrieval misses
- species structure is metric

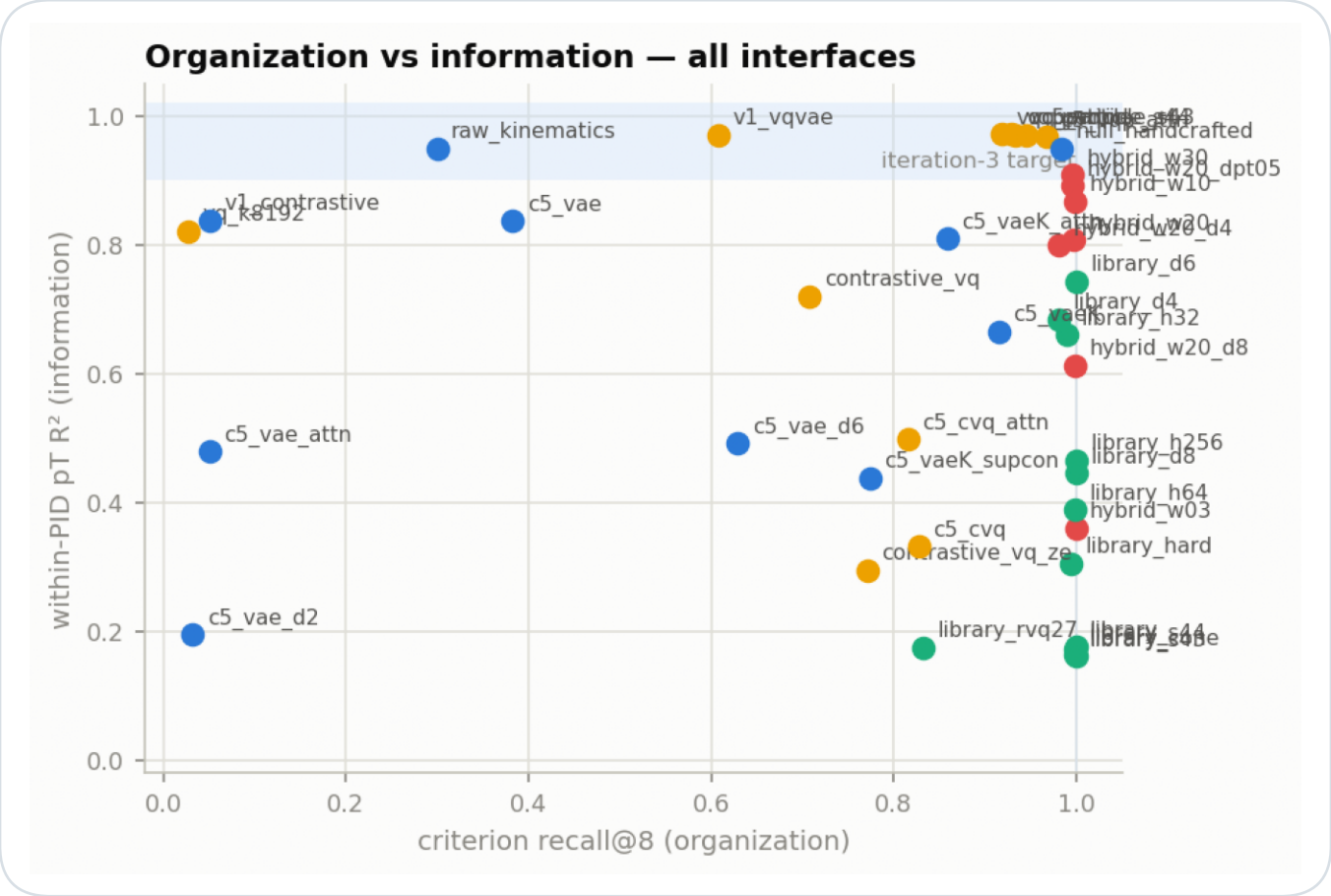
RECONSTRUCTION VQ

owns the information

- within-PID pT R^2 0.97
- PID / charge linearly decodable
- 27 bits, faithful

Geometry and information are two axes — and there's no downstream cost: library tokens probe at the raw-feature ceiling (0.744 acc / 0.931 AUC).

CAMPAIGN 3 · THE FRONTIER



Organization vs information. The hybrid weight traces the frontier toward the shaded 'both at once' corner but never reaches it — so recon_weight is a release dial, not a solved optimum.

CAMPAIGN 3 · A FREE WIN

The token halves for free

16-D → 8-D

Compressing the token raised criterion recall (0.9968 → 0.9982) and PID purity (0.918 → 0.987). The 16-D token was larger than its 5-column input; the criterion never needed the extra dimensions.

CAMPAIGN 4 · ROBUST NEGATIVE

Don't quantize the criterion — shrink the token

CRITERION-SHAPED VQ

fails

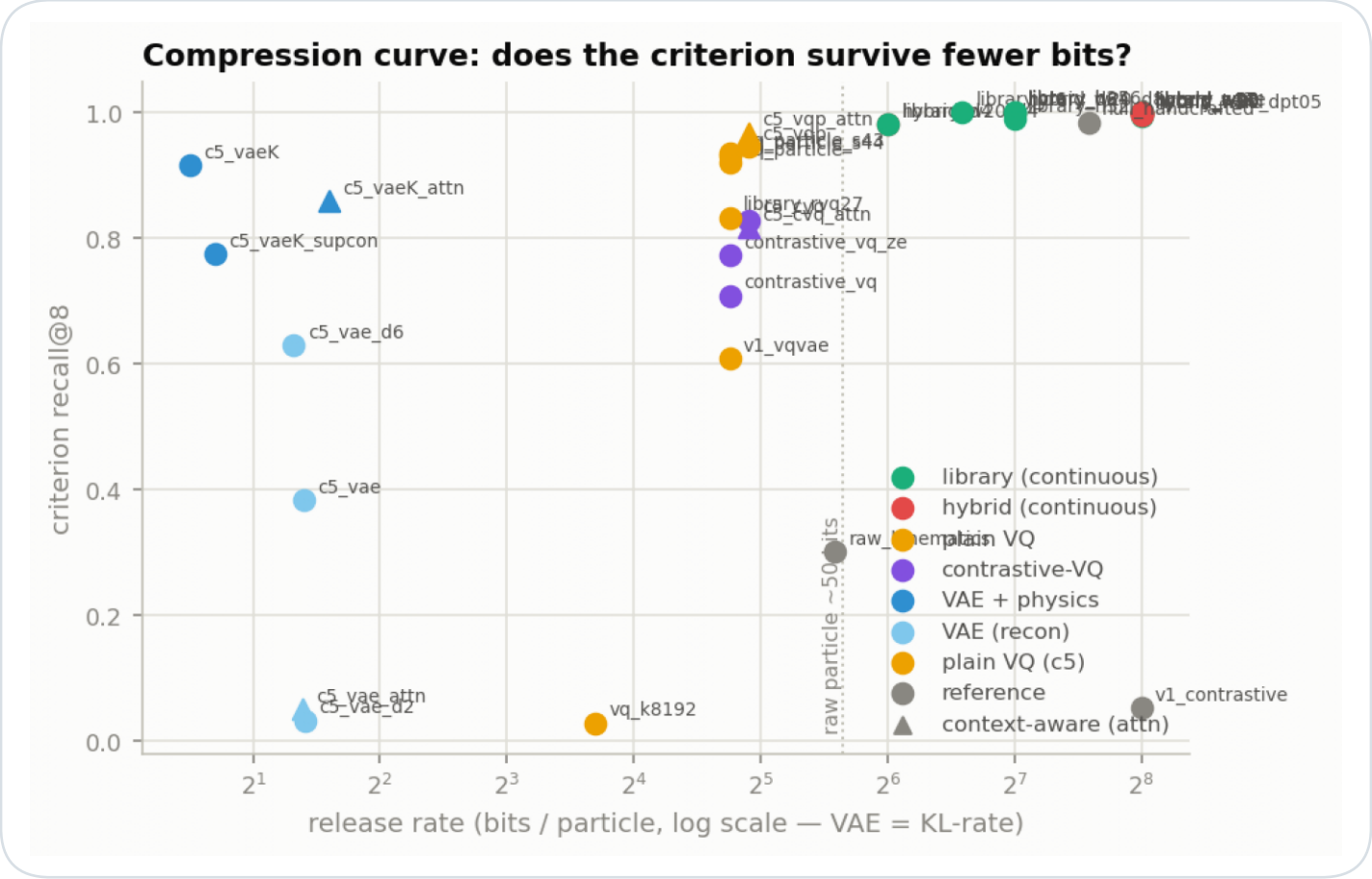
- recall 0.772 vs plain VQ 0.927
- kernel spreads to 15+ dims
- quantization can't hold it

LOW-DIM CONTINUOUS

works

- dim-6 = 96 bits, recall 0.9996
- pT R^2 rises 0.18 → 0.74
- granularity beats bit-count

CAMPAIGN 4 · COMPRESSION



Recall@8 vs release rate (bits/particle, log scale). Every token on one axis; the dotted line is the ~50-bit raw particle. Triangles mark the context-aware attention encoder.

CAMPAIGN 5 · A CLEAN 2×2×2

Put every choice on one axis

Bottleneck {VAE continuous · VQ discrete} × objective {reconstruction · + physics term} × encoder {context-free MLP · context-aware attention} — everything else matched, all on one bits axis (the VAE's rate read from its KL).

Which of the three choices actually decides the token?

CAMPAIGN 5 · THE ANSWER

Placement is decisive

PHYSICS TERM ON VAE LATENT

0.383 → 0.916

smooth mean → clean gradient → it works

SAME TERM ON VQ CODEBOOK

0.945 → 0.827

straight-through + EMA → it hurts

Same term, opposite sign. And the contrastive VAE reaches recall 0.916 at ~1.4 bits — sub-raw and criterion-strong.

CAMPAIGN 5 · CONTEXT

Attention is an asymmetric trade

LOW-RATE VAE + ATTENTION

0.383 → 0.051

context crowds out the criterion

HIGH-RATE VQ + ATTENTION

0.945 → 0.967

it can afford both

Context buys jet information — but folds the jet environment into the per-particle code.
Only a high-rate token affords both context and criterion.

SYNTHESIS

What five campaigns establish

01 **State the metric explicitly**
physics criterion beats an implicit augmentation metric.

02 **Keep the encoder context-free**
the same particle, the same token, everywhere.

03 **Geometry ≠ information**
a controllable frontier, not one optimum.

04 **Compress the continuous token**
don't force the criterion through a codebook.

05 **Placement and rate decide**
not the architecture.

FRONTIER

Where this points next

DATA-BLOCKED

Cross-experiment invariance

CMS + ATLAS in mixed batches — the north star; needs a second-detector sample.

RUNS TODAY

Downstream FM probe · rate-matched VAE ↔ VQ

Train a backbone on the exported tokens; β-sweep the VAE to a 30-bit point.

The token to release is a stated choice along sub-raw · faithful · criterion — which is exactly why the criterion is written down.